

Hypothesis Testing Summary

ApexPlanet Data Analytics Internship – Task 4
Data-Driven Sales and Customer Insights
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1. Objective

The purpose of this hypothesis test was to examine whether there is a statistically significant linear relationship between Quantity and Total Sales in the sales dataset.

2. Hypotheses

H0 (Null Hypothesis): There is no statistically significant linear relationship between Quantity and Total Sales.

H1 (Alternative Hypothesis): There is a statistically significant linear relationship between Quantity and Total Sales.

3. Method Used

The Pearson correlation test was performed using Python and the SciPy library. The analysis used 1,000 observations from the Task 3 dashboard dataset.

4. Test Results

Measure	Result
Sample size	1,000
Pearson correlation (r)	0.6466406163
P-value	$1.71360810896 \times 10^{-11}$
Significance level (α)	0.05
95% confidence interval for r	0.6091 to 0.6813

5. Decision

The p-value is far below the significance level of 0.05. Therefore, the null hypothesis is rejected.

6. Interpretation

The Pearson correlation of approximately 0.647 indicates a strong positive linear relationship between Quantity and Total Sales in this dataset. The 95% confidence interval for the correlation is approximately 0.609 to 0.681, which is entirely above zero.

This result provides strong statistical evidence of an association between the two variables. It does not, by itself, establish that changes in Quantity cause changes in Total Sales.

7. Business Relevance

The result supports the business observation that transactions involving higher quantities tend to have higher total sales. This can be considered alongside the other sales and customer findings when planning product, sales, and customer strategies.

8. Tools Used

Python, Pandas, SciPy, and Google Colab.

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